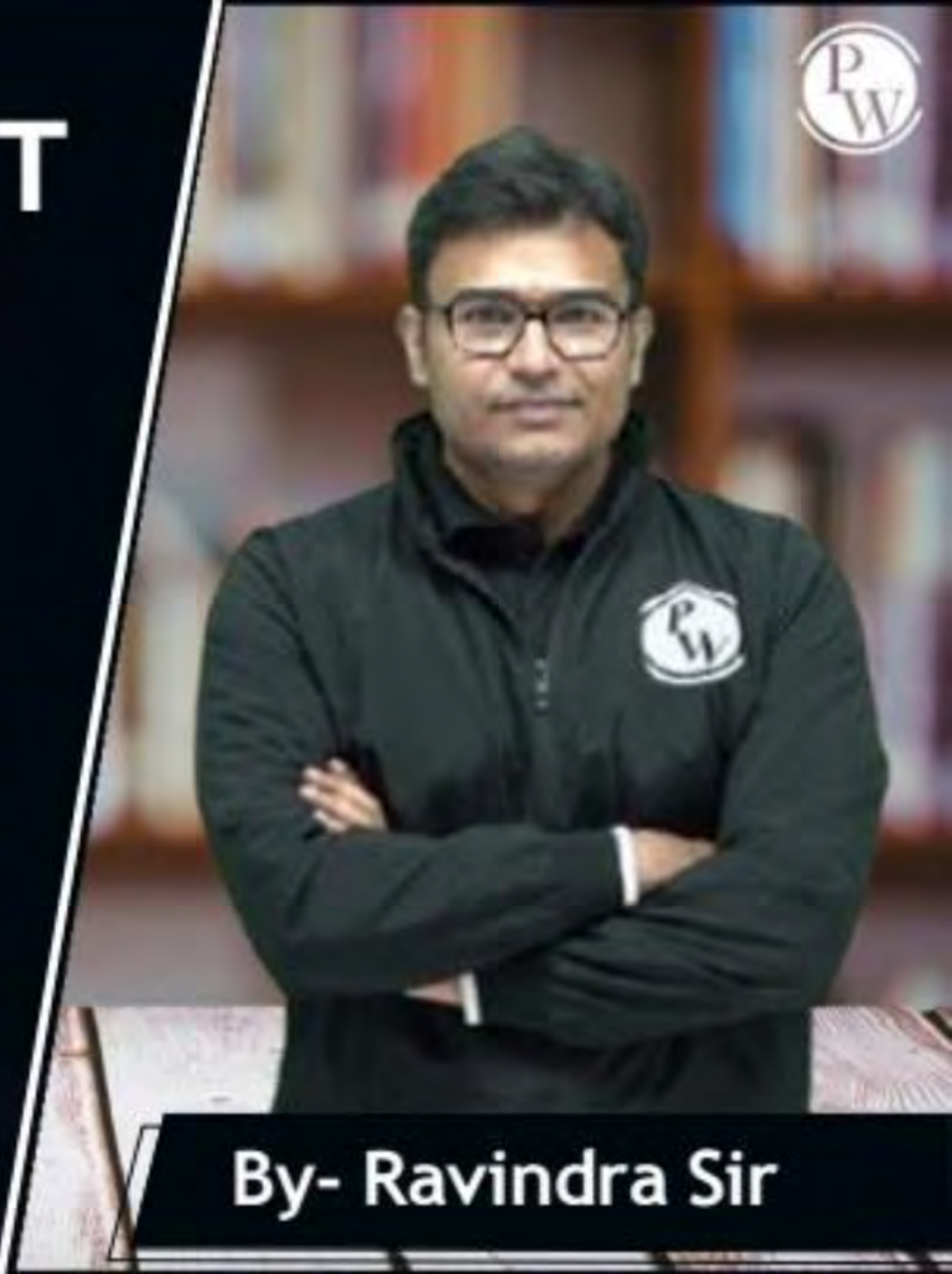
Computer Science & IT

COMPUTER NETWORK
(CN)

Application Layer Protocols

Lecture No. 04



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

NAT, PYQ's

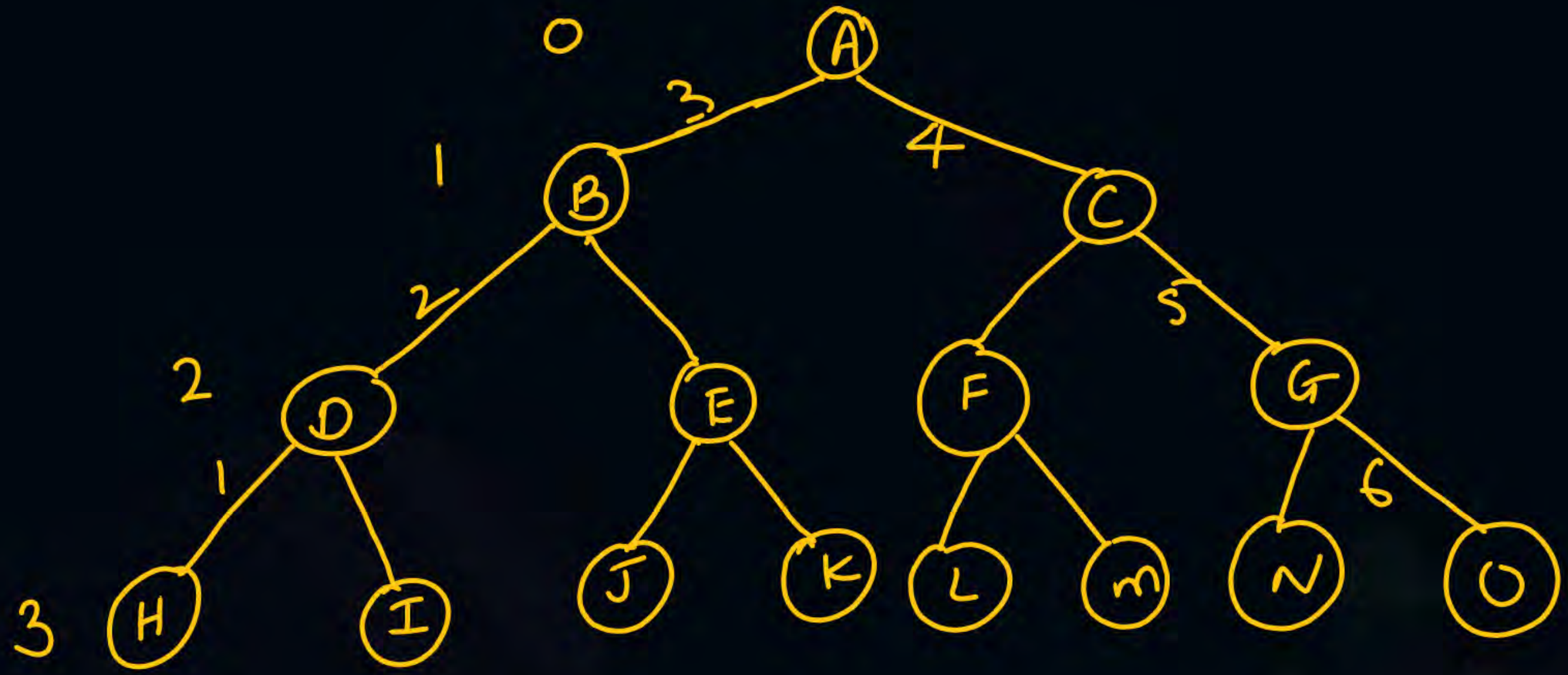
Topics to be Covered



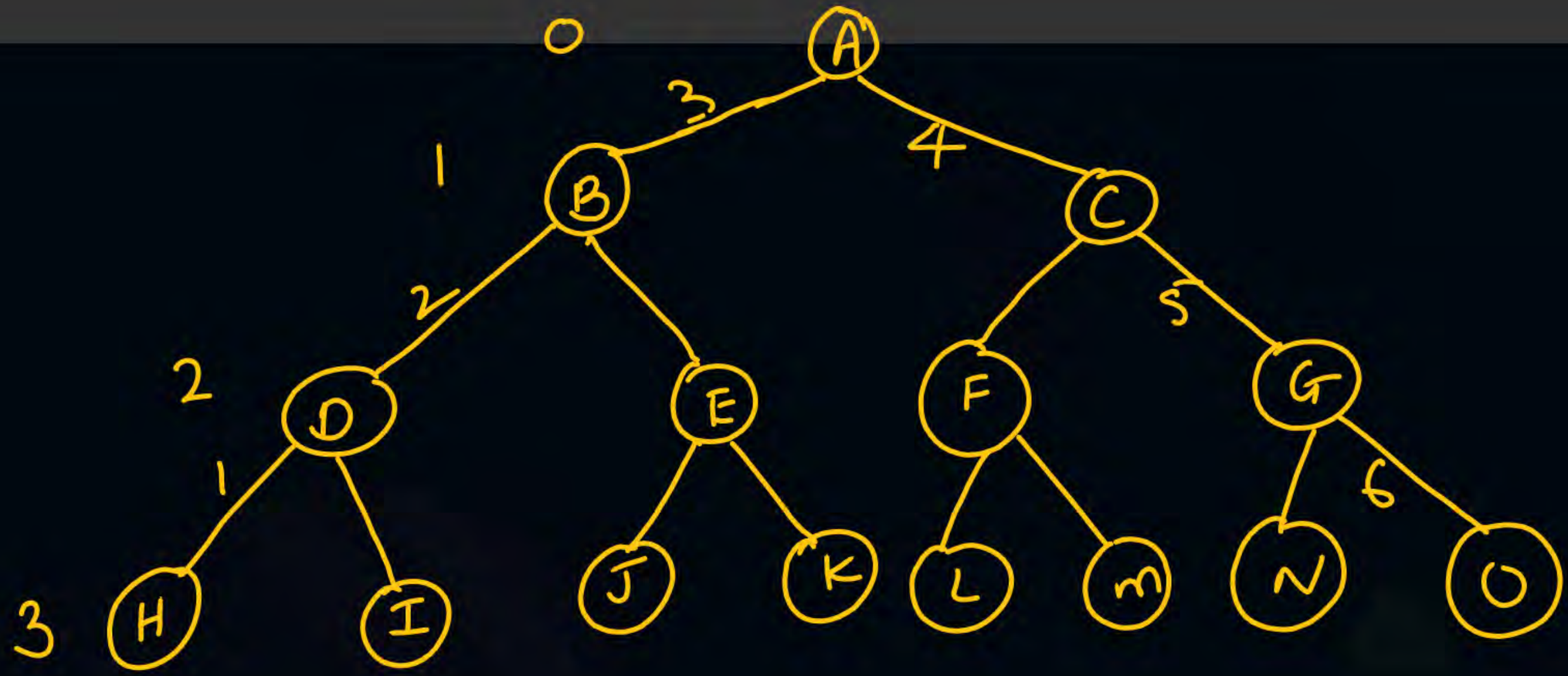
- Topic PYQ
- Topic
- Topic

→ A group of 15 routers are interconnected in a centralized complete binary tree with a router at each tree node. Router i communicates with router j by sending a message to the root of the tree. Then root sends the message down back to router j. The mean no of hops per message, assuming all possible routers are equally likely

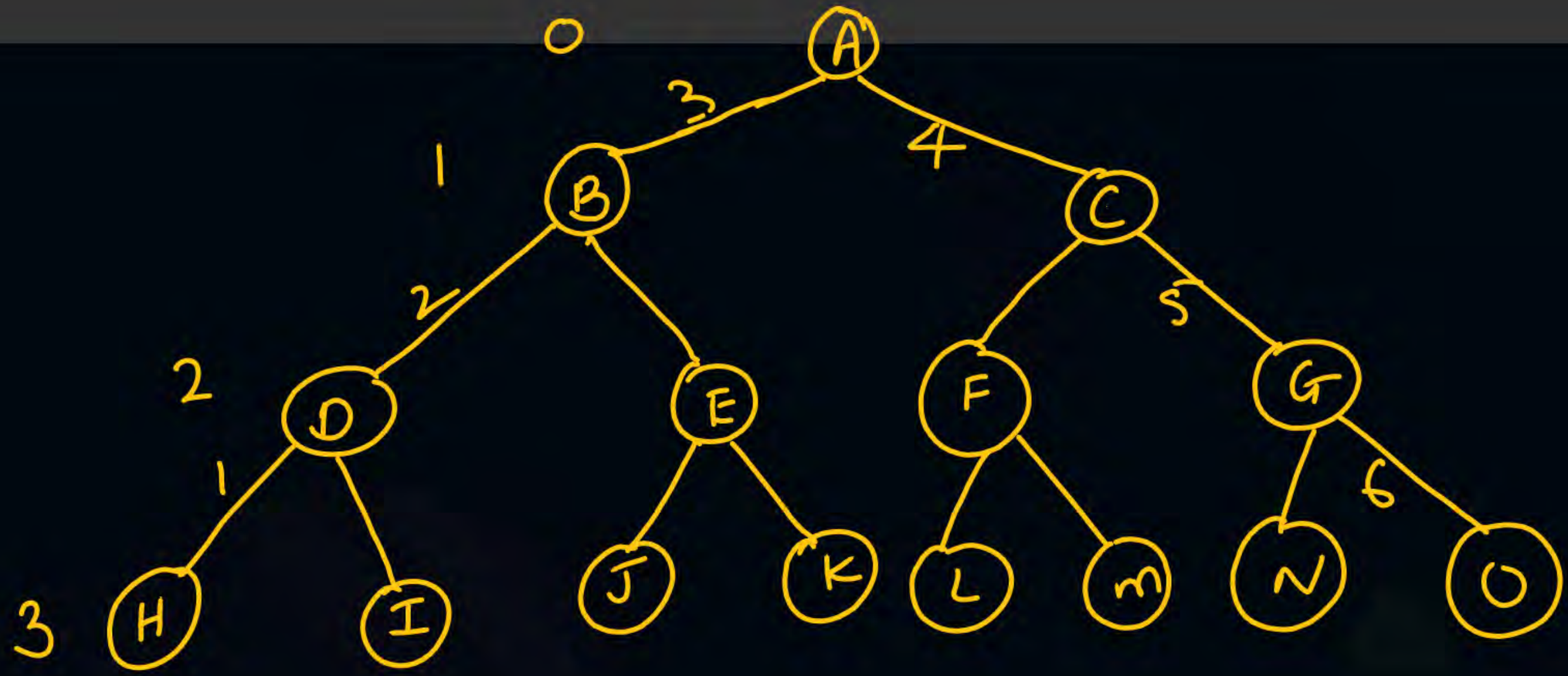
a) 3 b) 4.26 c) 4.53 d) 5.26



H – 3 level – 6 * 7
2 level – 5 * 4 = 73 X 8
1 level – 4 * 2
0 level – 3 * 1



Ans: 5×8
$4 \times 3 = 60 \times 4$
3×2
2×1

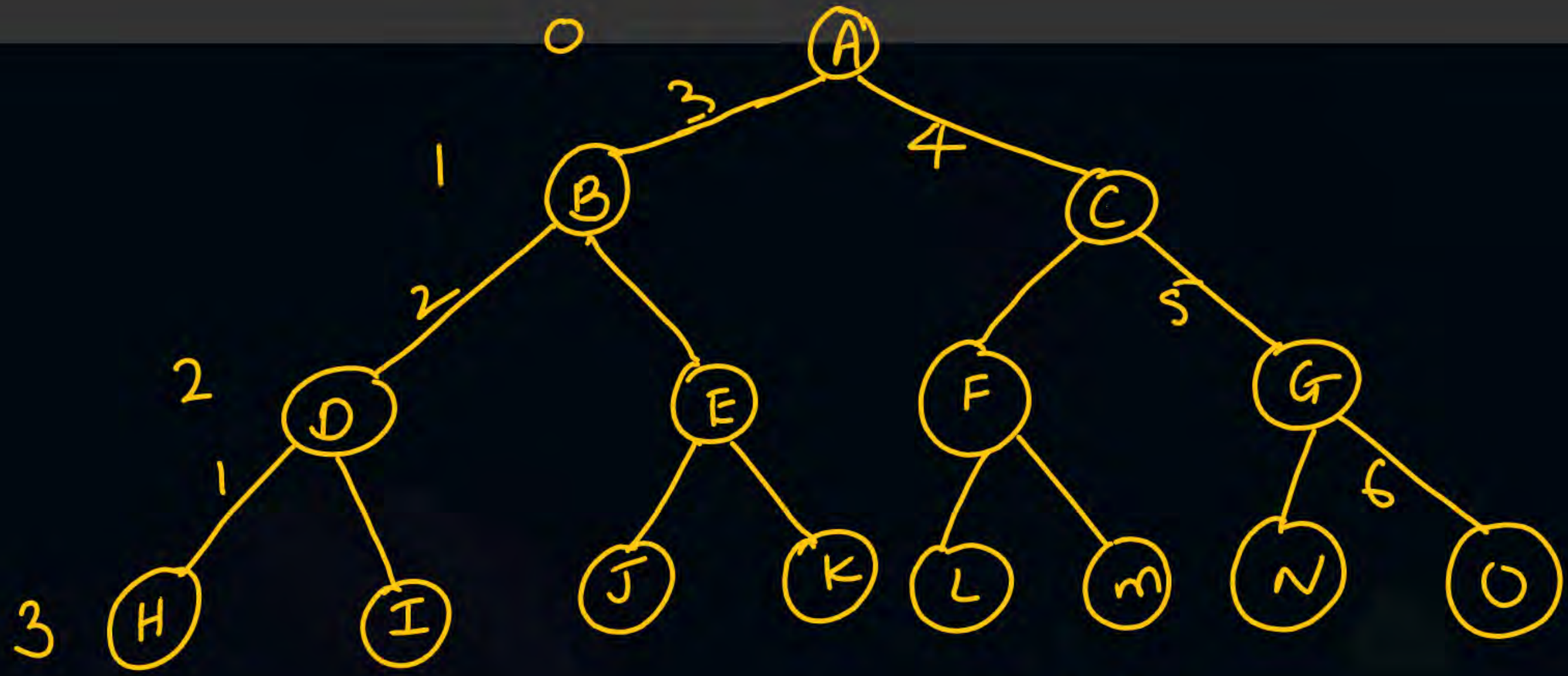


At B:- 4×8

$3 \times 4 = 47 \times 2$

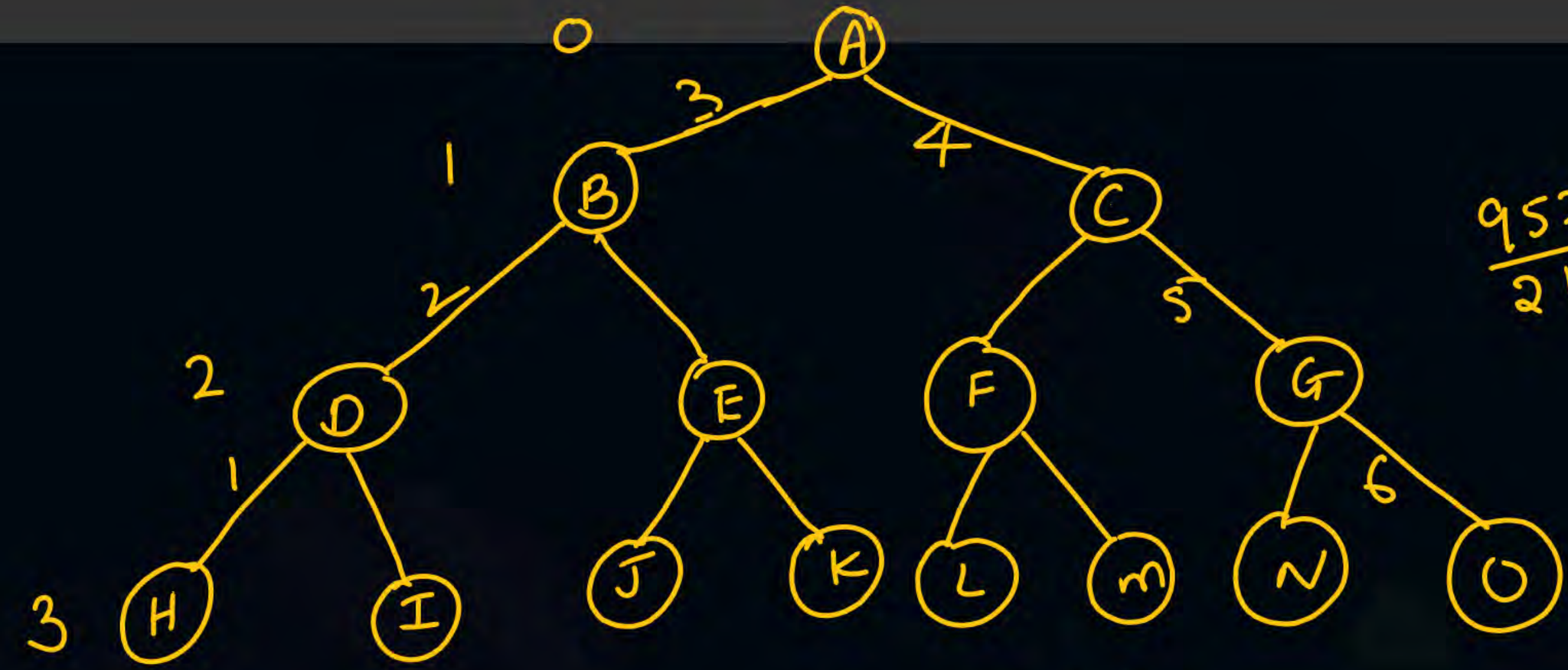
2×1

1×1



At A :- $3 \times 8 = 34$
2×4
1×2

Total no of hops
$= 584 + 240 + 94 + 34 = 952$



$\frac{952}{210} = 4.53$

Total no of messaye

$2 * {}^{15}C_2 = \underline{210}$

Total no of hops

$= 584 + 240 + 94 + 34 = \underline{952}$

⇒ Two popular routing algo are DVR and LSR. Which of the following statements are true

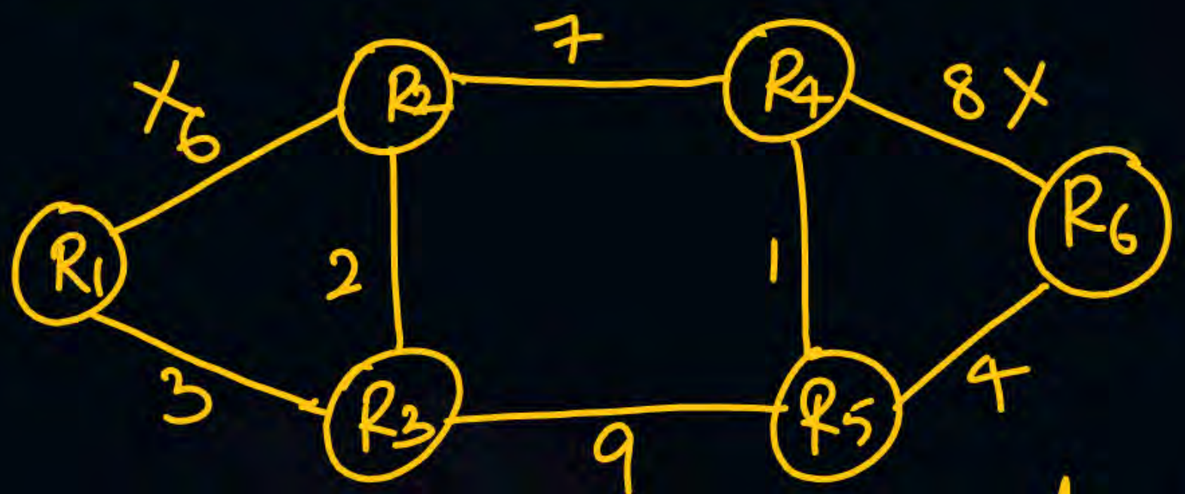
✓ S_1: Count to infinity is a problem only with DVR not with LSR

✗ S_2: In LSR, the shortest path algo is run only at one node

✗ S_3: In DVR, " " " " " "

✓ S_4: DVR requires less no of messages than LSR

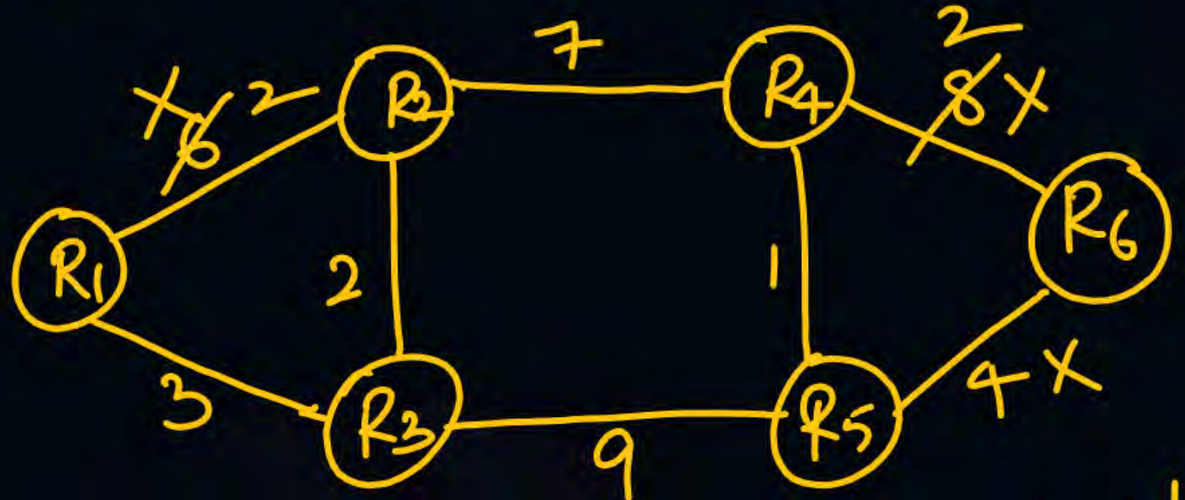
Consider a N/w with 6 Routers



Q) DVR is used. After all routing table stabilize, how many links will never be used for carrying data.

a) 4 b) 3 c) 2 d) 1

Consider a N/w with 6 Routers



Q) Suppose the weights of unused links are changed to 2 and PVR is used, How many links remain unused

a) 0 b) 1 c) 2 d) 3

→ Consider different activities related to email

m_1 : Send an email from a mail client to mail server

m_2 : Download an email from mail server to mail client.

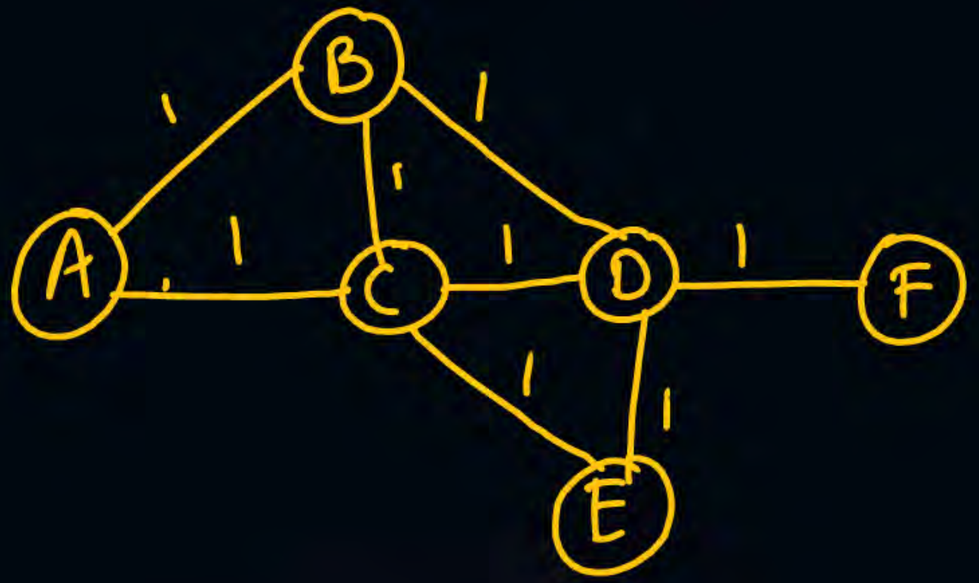
m_3 : checking email in the web browser

a) m_1 : HTTP m_2 : SMTP m_3 : POP

b) m_1 : SMTP m_2 : FTP m_3 : HTTP

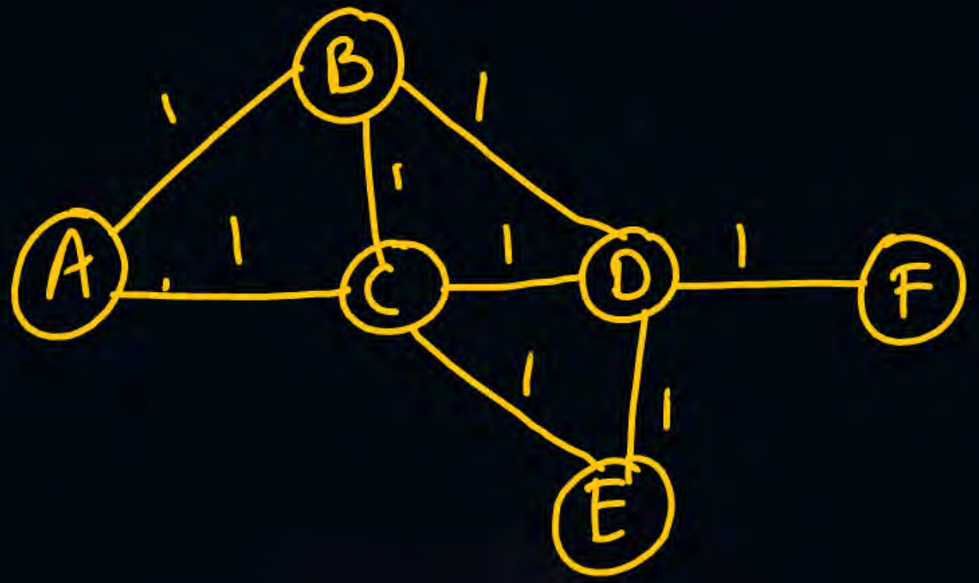
c) m_1 : SMTP m_2 : POP m_3 : HTTP

d) m_1 : POP m_2 : SMTP m_3 : IMAP



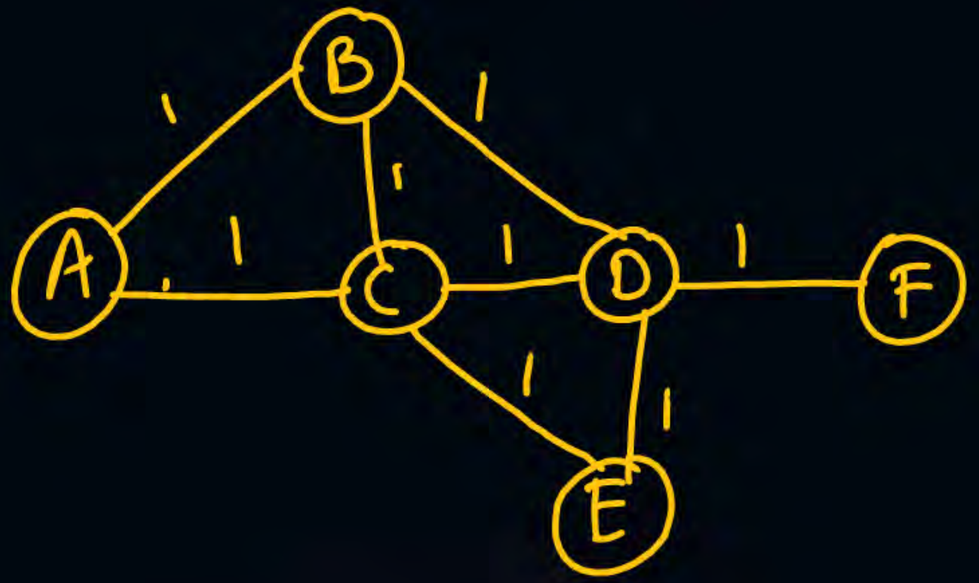
At :- A

Dest	Dist	NH
A	0	A
B	1	B
C	1	C
D	2	C
E	2	C
F	3	B



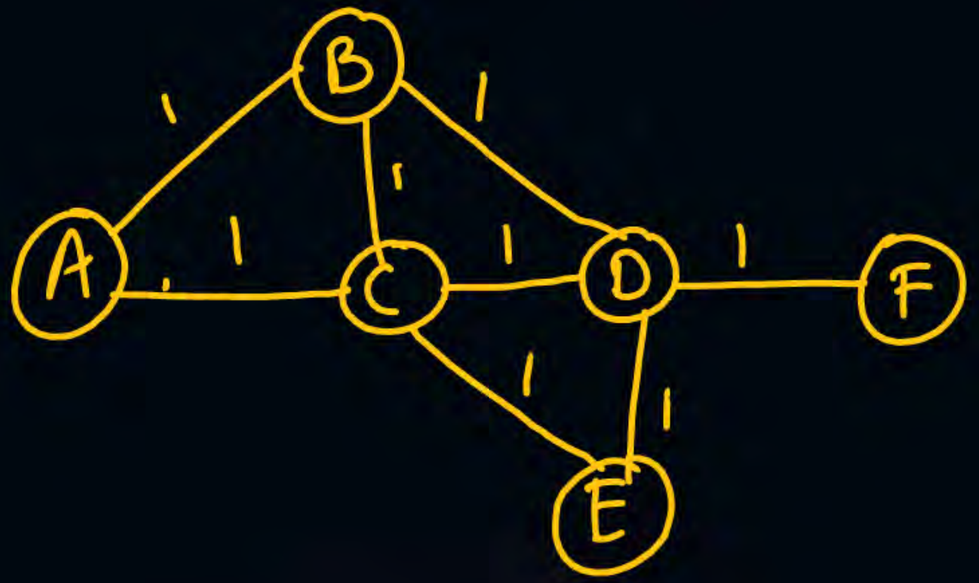
AtB :-

A	1	A
B	0	B
C	1	C
D	1	D
E	2	C
F	2	D



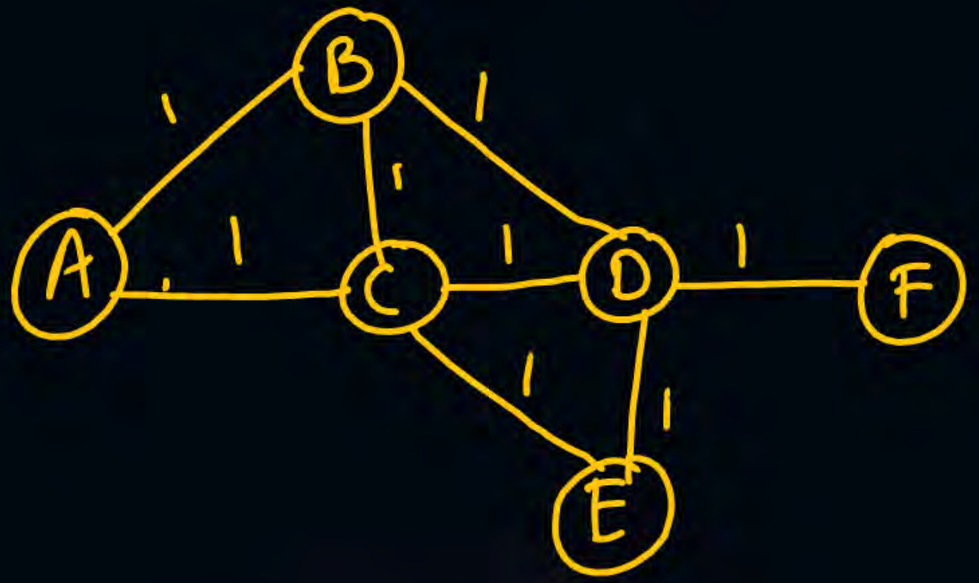
At C :-

A	1	A
B	1	B
C	0	C
D	1	D
E	1	E
F	2	D



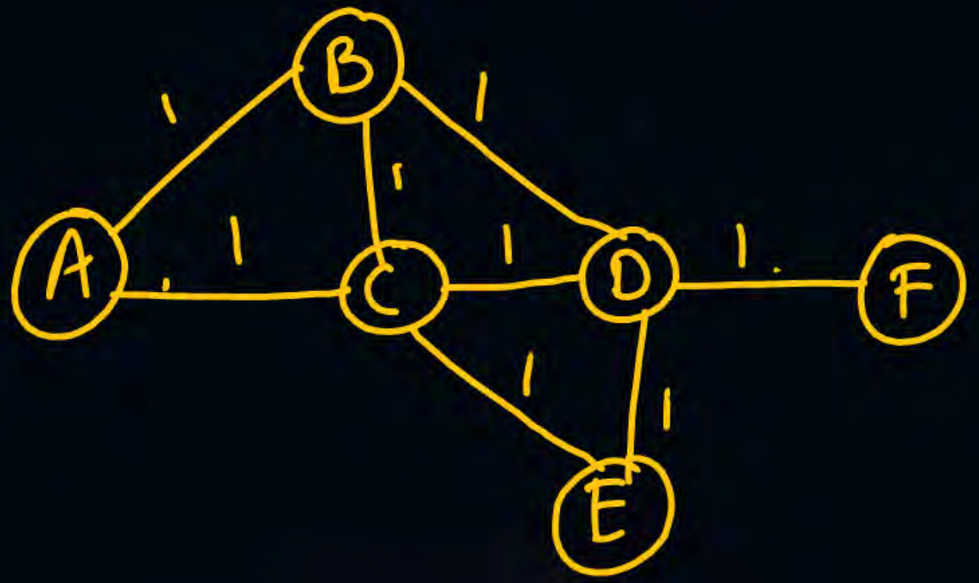
A to D :-

A	2	C
B	1	B
C	1	C
D	0	D
E	1	E
F	1	F



At E:-

A	2	C
B	2	C
C	1	C
D	1	D
E	0	E
F	2	D



At F:-

A	3	D
B	2	D
C	2	D
D	1	D
E	2	D
F	0	F

The protocol Data unit for the application layer in the Internet stack is

a) Segment b) Datagram c) message d) Frame

AL — message
TL — Segment
NL — Datagram
DLL — Frame
PL — IPDU

→ The transport layer protocols used for real time multimedia, file transfer, DNS and email respectively are

a) TCP, UDP, UDP and TCP

b) UDP, TCP, TCP and UDP

c) UDP, TCP, UDP and TCP

d) TCP, UDP, TCP and UDP

Assume that a Source S and destination D are connected through two routers. Determine how many times each Packet had to visit the NL and DLL during a transmission from S to D

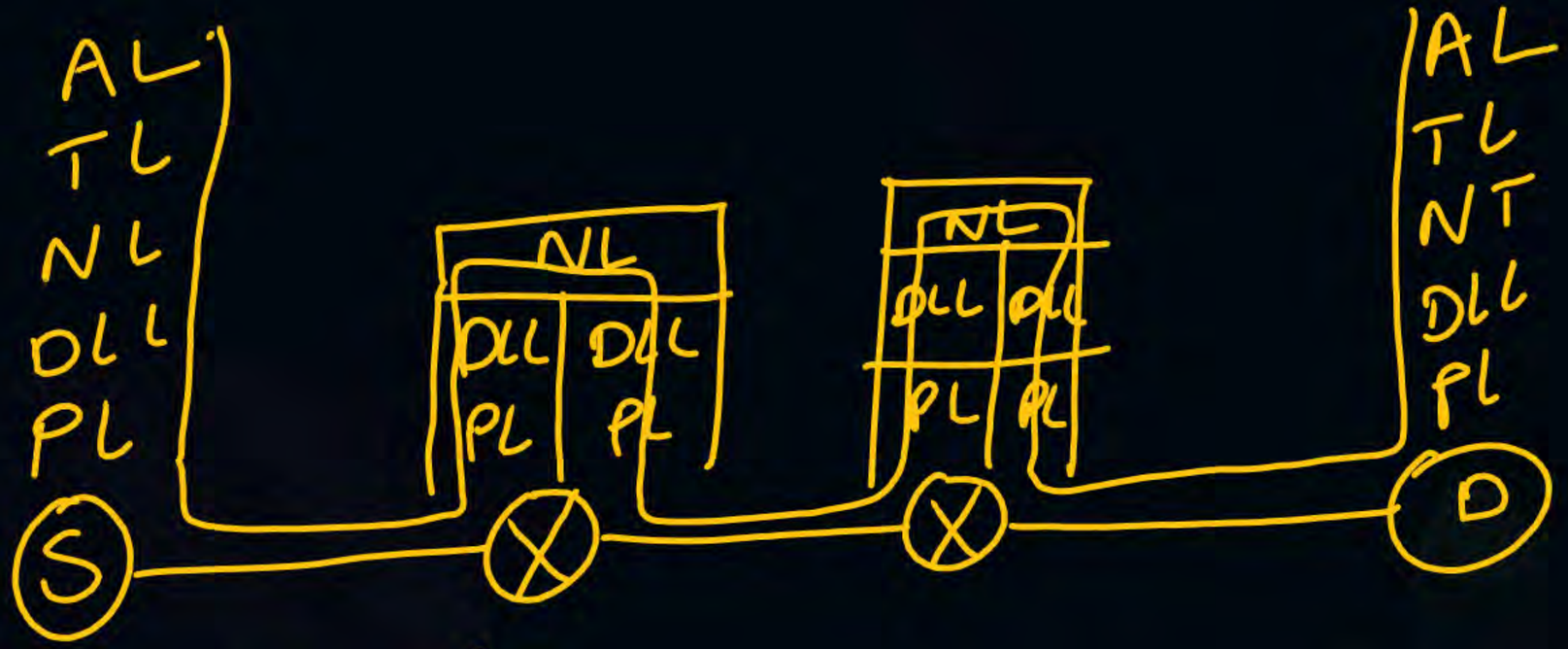


a) NL - 4, DLL-4

b) NL-4, DLL-3

c) NL-4, DLL-6

d) NL-2, DLL 6 times



NL - 4

DLL - 6

– Consider the following 3 statements about DVR and LSR for a large N/w with 500 Routers and 4000 links

✓ S_1: The Computational overhead in LSR is higher than DVR.

✗ S_2: The DVR (with split horizon) avoids persistent loops, but not LSR.

✓ S_3: After a topology change, LSR will converge faster than DVR.

Which of the following is true about RIP and OSPF

a) RIP uses DVR, OSPF uses LSR

b) OSPF uses DVR, RIP uses LSR

c) RIP, OSPF uses LSR

d) RIP, OSPF uses DVR

THANK - YOU